

CONDUCTORS UNDER COEFFICIENT CHANGE AND NONMONOMIAL HILBERT BOUNDARIES

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ABSTRACT. Let $T \rightarrow S$ be a normal proper birational model of a smooth variety in characteristic zero, let $X \rightarrow S$ be finite flat with X smooth, and let Z normalize $T \times_S X$. We express the actual conductor ideal on Z as the difference between the pulled-back Jacobian divisor of the coefficient map and the ramification divisor of $Z \rightarrow T$. The identity is compatible with towers of covers and includes conditions in every codimension. Applied to polynomial-contact Hilbert graphs, it determines conductors and normalization defects for all tangent covers $b = u^r$, $c = u + v^s$. For the square-tangency cover and every contact order, we determine the complete normalized surface and its scheme fibre: two rational double points are followed by an explicit chain of smooth point blow-ups. Equal embedded Hilbert limits can then have a regular normalization lift and a singular one. The self-contained diagonal specialization also recovers the full nonreduced parameter fibre and its exact nilpotence order.

1. INTRODUCTION

A flat embedded limit, the normalization of its parameter space, and the scheme fibre of that normalized space need not retain the same information. The conductor measures a further difference: it records precisely which functions on the normalization descend to the original parameter space. Computing its generic order on boundary divisors is not by itself a computation of the conductor at their intersections. This distinction is particularly relevant when a change of coefficients turns a smooth Hilbert graph into a nonnormal surface.

The first result of this paper gives an identity of actual conductor ideals under finite flat changes of a smooth coefficient space. It applies to any normal proper birational model, in any dimension. Its proof uses classical trace duality. The geometric applications then leave the diagonal toric setting: two coefficient divisors become tangent, distinct divisorial valuations can agree on the two original coordinate functions, and one Hilbert point can have normalization lifts with different singularity types.

1.1. The conductor identity. Let $h : X \rightarrow S$ be a finite flat dominant map of smooth varieties over a characteristic-zero field, and let $f : T \rightarrow S$ be proper birational with T normal. Write

$$Y = T \times_S X, \quad Z = Y^\nu, \quad \pi : Z \rightarrow X, \quad \psi : Z \rightarrow T.$$

The fibre product is integral. Let R_h be the Cartier divisor of $\text{Fitt}_0 \Omega_{X/S}$, and let R_ψ have coefficient $e_Q - 1$ at a prime divisor Q of Z , where e_Q is the ramification index over T . Theorem 2.1 gives

$$\mathfrak{c}_{Z/Y} = \mathcal{O}_Z(R_\psi - \pi^* R_h).$$

This is not just an isomorphism of rank-one modules or an equality on an open set containing the generic points of the divisors. It identifies the conductor as a reflexive ideal at every point. In particular its order at Q is $v_Q(\text{Fitt}_0 \Omega_{X/S}) - e_Q + 1$.

The resulting defect cycle is twice the generic length of the normalization quotient, weighted by residue degrees (Corollary 2.4). The conductor divisors add under towers of finite flat covers

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in the precise sense of Theorem 2.5; this involves reflexive products, not an unproved equality of ordinary tensor products. No smoothness or Cohen–Macaulay assumption on T beyond normality is needed for the conductor identity. The normalization of a rational Hilbert graph is therefore an immediate source of examples.

1.2. Tangent polynomial contacts. For $a \geq 2$ consider the marked family of embedded graphs of

$$[s_0 : t_0] \mapsto [t_0^a : bs_0^a : cs_0t_0^{a-1}] \quad \text{in } \mathbb{P}^1 \times \mathbb{P}^2.$$

We prove directly that its retained-coefficient Hilbert graph is the smooth chain blow-up of $\prod_{j=1}^a(b, c^j)$. This proof gives a finite row factorization of the evaluation ideal, removes its principal content, and derives an exhaustive affine cover. It uses no assertion about an unmarked finite algebra and no theorem from the reconstruction paper.

Now change coefficients by $b = u^r$, $c = u + v^s$, with $r, s \geq 2$. The conductor identity determines every divisorial conductor order and generic defect for every a, r, s ; the residue extensions are computed rather than inferred from the number of roots in a closed fibre (Theorem 4.1). For $r = s = 2$ we compute the complete normal surface in every contact order (Theorem 4.2). At order two its middle chart is

$$k[e, z, v]/(v^2 - ez(z - 1)), \quad u = ez,$$

with conductor (e) and scheme fibre $k[e, z]/(ez)$. The two points $z = 0, 1$ on that chart are rational double points. Each higher contact order adds one ordinary blow-up at a smooth endpoint. The full parameter fibre stays a reduced nodal chain, although the conductor orders grow along it.

The first two exceptional divisors both have values $(2, 1)$ on (u, v) , but their values on $u + v^2$ are 4 and 2. Thus the result cannot be read from a fan in the original two monomial weights. Moreover the Hilbert coordinate is z^2 along the second divisor. The points $z = 1$ and $z = -1$ give the same embedded Hilbert limit, while one normalization germ is singular and the other is regular. Explicit projective coefficient arcs realize both points (Corollary 4.3).

1.3. Relation to classical results and scope. The trace complementary module and the equality of the Kähler and Dedekind differentials for finite flat complete intersections are classical; precise formulations appear in [1, Sections 49.8, 49.12 and 49.15]. Our proof of the conductor identity exposes this dependence. It is not a claim to a new duality formalism. The Newton-polyhedron criterion, normalized Rees constructions and the complete-ideal theory used in the marked chain belong to the established theory of integral closure [2, Chapters 1, 5 and 14]. The same reference states Jacobian containment results and the Lipman–Sathaye theorem in its conductor chapter. Here the stronger geometric hypotheses permit an equality of the actual conductor with a specified complementary module, followed by the explicit nonmonomial normalization and fibre calculations above. No exhaustive priority claim for the general identity is made.

The diagonal family $b = u^\rho$, $c = v^\sigma$ is treated separately in Section 5. There the full coordinate ideal on a proper toric cone is divisorial; a proof on the character lattice shows that the scheme fibre has no embedded points. The reduction at a cyclic quotient crossing is computed as an ordinary node, and ordinary powers give the exact nilpotence order. The conductor is then recovered from the same coefficient-change identity, including an explicit unequal-cover chart and overlap generators.

The conductor theorem is not restricted to these two surface families. The complete surface and fibre classifications, however, refer to the stated families. Neither the conductor formula nor the examples assert a classification of all higher-corank or singular quadratic-pencil Hilbert fibres. Normalizing the parameters does not normalize the embedded universal curves. All uses of a finite failure algebra require the separate effective reconstruction hypotheses; none enters

the proofs in this paper. The accompanying technical supplement retains the earlier power-ideal, equivariant, contact and reconstruction-application arguments for reference, without making them prerequisites of the present central route.

2. CONDUCTORS UNDER FINITE FLAT CHANGE OF COEFFICIENTS

Throughout this section k is a field of characteristic zero. A variety is an integral separated scheme of finite type over k . If V is normal and $D = \sum d_Q Q$ is a Weil divisor, $\mathcal{O}_V(D)$ denotes the subsheaf of its function field specified by $v_Q(z) + d_Q \geq 0$ at every prime divisor. Products of such sheaves are ordinary products unless a double dual is explicitly written. For a finite dominant morphism of normal varieties, pullback of a Weil divisor means its valuation pullback, with the ramification indices as coefficients; no Cartier assumption is implicit.

Let $h : X \rightarrow S$ be a finite flat dominant morphism of smooth varieties, and let $f : T \rightarrow S$ be a proper birational morphism with T normal. There is no restriction on their common dimension. Set

$$Y = T \times_S X, \quad Z = Y^\nu, \quad \nu : Z \rightarrow Y, \quad \pi : Z \rightarrow X, \quad \psi : Z \rightarrow T.$$

The Jacobian ideal $\mathfrak{j}_h = \text{Fitt}_0 \Omega_{X/S}$ is invertible and nonzero. Write $\mathfrak{j}_h = \mathcal{O}_X(-R_h)$. For a prime divisor Q of Z over a prime divisor P of T , let e_Q be its ramification index, and put $R_\psi = \sum_Q (e_Q - 1)Q$. These definitions use the morphisms, not choices of local coordinates.

Theorem 2.1 (Conductor and coefficient change). *In the preceding situation, Y is integral and S_2 , and its normalization is finite. The conductor, as an actual ideal of $\mathcal{O}_Z$, is*

$$\mathfrak{c}_{Z/Y} = \mathcal{O}_Z(R_\psi - \pi^* R_h). \quad (2.1)$$

In particular it is reflexive, and its coefficient at Q is

$$c_Q = v_Q(\mathfrak{j}_h) - e_Q + 1 \geq 0. \quad (2.2)$$

There is no further condition supported in codimension at least two. The scheme Y is normal if and only if $c_Q = 0$ for every Q . The conductor is supported over the locus on which f is not an isomorphism. If T is smooth, Y is moreover a local complete intersection and hence Cohen–Macaulay.

The identity is an application of classical complementary modules and finite duality, not a new duality theorem. Its use here is that the same formula computes a global conductor for nonmonomial coefficient changes, for singular normal models, and for models in higher dimension. The explicit tangency calculations in Section 4 will supply its first nonmonomial Hilbert-boundary application.

2.1. Trace lattices and the actual conductor ideal. We give the algebraic argument, including the identification as an ideal rather than just an isomorphism of abstract modules. For a normal Noetherian domain A with fraction field K , a finite separable extension L/K , and a finite A -subalgebra $E \subset L$, define

$$E^\dagger = \{z \in L : \text{Tr}_{L/K}(zE) \subset A\}.$$

The trace pairing identifies this with $\text{Hom}_A(E, A)$. This is the Dedekind complementary module [1, Section 49.8]. We shall also use the classical equality of the different and the Kähler different for a finite flat local complete intersection map [1, Section 49.12].

Lemma 2.2 (A trace-lattice identity). *Suppose E is finite locally free over A , is a domain with fraction field L , and $E^\dagger = j^{-1}E$ for a nonzero $j \in E$. Let B be the finite normalization of E in L . Then, inside L ,*

$$(E : B) = jB^\dagger. \quad (2.3)$$

Here $(E : B) = \{z \in L : zB \subset E\}$ is contained in B and is the ordinary conductor. The assertion is local on $\text{Spec } E$ when $E^\dagger$ is only locally generated in this way.

Proof. For every E -module M there is the elementary adjunction

$$\text{Hom}_E(M, \text{Hom}_A(E, A)) \simeq \text{Hom}_A(M, A).$$

The forward map evaluates at $1 \in E$; the inverse sends a functional λ to $m \mapsto (e \mapsto \lambda(em))$. Apply it to $M = B$. After extension to K , all functionals are given uniquely by the trace pairing on the field L . Thus the adjunction identifies the two submodules of L by the same multiplier, not by an unspecified unit:

$$B^\dagger = \{z \in L : zB \subset j^{-1}E\} = j^{-1}(E : B).$$

Since $1 \in B$, a conductor multiplier lies in $E \subset B$. Conversely $(E : B)$ is a B -ideal. This proves the claimed equality as ideals. Localization and change of a local generator multiply both sides by the same unit, so the local identities glue. $\square$

Lemma 2.3 (The complementary module on a normal model). *Let $A \subset B$ be a finite inclusion of normal domains essentially of finite type over a characteristic-zero field, with separable fraction fields $K \subset L$. On $\text{Spec } B$ one has*

$$B^\dagger = \mathcal{O}_B \left(\sum_Q (e_Q - 1)Q \right),$$

where Q runs over prime divisors and e_Q is computed over its height-one contraction to A .

Proof. A normal domain is the intersection of its height-one localizations in its fraction field. Hence $z \in B^\dagger$ if and only if the trace of zb is in A_P for every height-one $P \subset A$ and every $b \in B$. The ring $B \otimes_A A_P$ is the semilocal normalization of the DVR A_P in L . Its maximal ideals are the primes Q over P . The local complementary module has valuation $-(e_Q - 1)$ at Q : residue characteristic zero gives tame ramification and a separable residue extension. This is the inverse-different formula for DVRs. One may see the exponent after an unramified extension by writing a uniformizer of A_P as a unit times an e_Q th power; the derivative has order $e_Q - 1$. Unramified trace duality contributes order zero.

Going down over the normal domain A , together with incomparability for integral extensions, shows that every height-one prime of B contracts to a height-one prime of A . Thus all height-one primes of B are included. Intersecting the preceding valuation conditions in L is exactly the displayed divisorial sheaf. This also proves its reflexivity without assuming that B is Cohen–Macaulay. $\square$

Proof of Theorem 2.1. On an affine open of T , the coordinate algebra of Y is finite locally free over the coordinate domain of T . It therefore injects into its generic algebra. Birationality of f identifies that generic algebra with the single field $k(X)$, not a product of fields. Consequently every such algebra is a domain. This proves integrality, including the exclusion of vertical or embedded structure. The finite flat morphism h is a local complete intersection: it is a morphism between smooth varieties of equal dimension. Its base change is syntomic of relative dimension zero. The flat depth formula, with zero-dimensional complete-intersection fibres, transfers S_2 from T to Y . If T is smooth the same description makes Y a local complete intersection. Finiteness of normalization follows from excellence.

Let C be the finite locally free algebra of X over an affine part of S , and A the coordinate algebra of an affine part of T above it. Put $E = C \otimes A$. Finite locally free duality and its trace element commute with arbitrary base change: this is checked by dualizing a finite free module.

The complete-intersection different formula therefore identifies the image of the trace in the invertible relative dualizing module with $j_h E$.

We may now localize A at a point of T . The finite algebra E is semilocal, so this invertible module and the pulled-back Jacobian ideal are free. Choose a generator j of $j_h E$. Under the trace identification the preceding assertion is exactly $E^\dagger = j^{-1} E$. This argument does not require a single global Jacobian equation on X , or apply a finite trace to a nonfinite localization of C .

Apply Lemma 2.2, and then Lemma 2.3, to the normalization B . They give

$$\mathfrak{c}_{Z/Y} = j \mathcal{O}_Z(R_\psi) = \mathcal{O}_Z(R_\psi - \pi^* R_h).$$

The ideal on the left is contained in $\mathcal{O}_Z$, so all its valuation orders are nonnegative. This is an equality of sheaves, and therefore already includes every point of codimension at least two. It is the unit ideal precisely when $B = E$. On an open where $T = S$ the base change is X , which is normal, so no conductor occurs there. All claims follow. $\square$

2.2. Defect cycles and composition. Write $C(Z/Y) = \pi^* R_h - R_\psi$ for the effective conductor divisor. Let $g : Y \rightarrow T$ be the finite projection. For a prime divisor $P \subset T$, let

$$\delta_P = \text{length}_{\mathcal{O}_{T,P}}((\psi_* \mathcal{O}_Z)/(g_* \mathcal{O}_Y))_P.$$

This is a length on the base DVR; it is not the length at an individual normalization branch.

Corollary 2.4 (Normalization defect). *In the situation of Theorem 2.1,*

$$2\delta_P = \sum_{Q|P} [k(Q) : k(P)] c_Q. \quad (2.4)$$

Thus $2 \sum_P \delta_P P = \psi_* C(Z/Y)$. The generic defect cycle is supported on the exceptional locus of $T \rightarrow S$.

Proof. Localize at P . Both the order E and its normalization B are free lattices of the same rank over the DVR $A = \mathcal{O}_{T,P}$. If a basis of E is expressed in a basis of B by a matrix M , then $v_A(\det M) = \text{length}_A(B/E) = \delta_P$. The matrices of the trace pairing satisfy $G_E = M^t G_B M$. Therefore

$$2\delta_P = v_A(\text{disc } E) - v_A(\text{disc } B).$$

The first discriminant is the norm of the different generator j_h , so its order is $\sum_{Q|P} [k(Q) : k(P)] v_Q(j_h)$. The second has order $\sum_{Q|P} [k(Q) : k(P)] (e_Q - 1)$ by tame DVR duality. Subtraction proves the formula. This argument also verifies that the right side is even; no geometric splitting assumption was used. $\square$

Theorem 2.5 (Transitivity of conductor divisors). *Let $X_2 \xrightarrow{h_2} X_1 \xrightarrow{h_1} S$ be finite flat dominant maps of smooth varieties. For a fixed normal proper birational model $T \rightarrow S$, let*

$$Z_i = (T \times_S X_i)^\nu, \quad \chi : Z_2 \longrightarrow Z_1, \quad W = Z_1 \times_{X_1} X_2.$$

Then W is integral, $Z_2 = W^\nu$ canonically, and

$$C(Z_2/T \times_S X_2) = \chi^* C(Z_1/T \times_S X_1) + C(Z_2/W). \quad (2.5)$$

These identifications are associative for successive finite flat covers. In terms of ideals, the equality is the reflexive product of the two conductor ideals after reflexive pullback; it need not be an equality of ordinary tensor products.

Proof. Finite flatness injects the coordinate rings of W into their generic field $k(X_2)$, so W is integral. The finite normalization map to $T \times_S X_1$ remains finite after base change. The two resulting integral rings have the same fraction field and one is integral over the other; their integral closures therefore coincide. This proves the canonical identification and finiteness of χ .

The chain rule gives $R_{h_1 h_2} = h_2^* R_{h_1} + R_{h_2}$. For a prime of Z_2 above Z_1 and T , ramification indices multiply; thus $e_1 e_2 - 1 = e_2(e_1 - 1) + (e_2 - 1)$ gives $R_{\psi_1 \chi} = \chi^* R_{\psi_1} + R_\chi$. Apply Theorem 2.1 to the original model and also to the normal model $Z_1 \rightarrow X_1$. Subtracting the two divisor identities gives (2.5). The second summand is an actual effective conductor divisor even when Z_1 is singular; this is why normal, rather than only smooth, models were allowed. Integral closure in one field and the chain rule are associative. Equality in codimension one gives the stated reflexive-product interpretation, and asserts no equality for unsaturated products. $\square$

Corollary 2.6 (Normal Hilbert models). *Let S be smooth and let a rational map from S to a separated Hilbert scheme be defined on a dense open. Let T be the normalization of its retained-coefficient graph closure, assumed proper over S . Every finite flat dominant map $X \rightarrow S$ from a smooth variety has the conductor and defect formulas above for $T \times_S X$ and its normalization, compatibly with towers of covers. The pulled-back universal embedded family remains flat.*

Proof. The normalized graph is a normal proper birational model of S , so the theorem applies directly. Flatness of the universal family is preserved by every base change, including normalization of parameters. If the original graph was not normal, $T \times_S X$ is the pullback of its normalization, not silently the original graph. The result computes the new conductor defect; it does not classify all fibres of that Hilbert scheme or normalize the embedded universal curves. $\square$

3. THE MARKED HILBERT GRAPH

We give the chain construction in full, so the applications require no unpublished predecessor theorem. Fix $a \geq 2$, let $S_a = \text{Spec } k[b, c]$, and consider

$$[s_0 : t_0] \mapsto [t_0^a : bs_0^a : cs_0 t_0^{a-1}]. \quad (3.1)$$

Over $b \neq 0$ this is a morphism, and its graph is an embedded curve in $X_0 = \mathbb{P}^1 \times \mathbb{P}^2$. Its Hilbert polynomial with respect to $\mathcal{O}_{X_0}(1, 1)$ is $P_a(l) = (a+1)l + 1$. Define T_a to be the schematic closure of the resulting section in $S_a \times \text{Hilb}^{P_a}(X_0)$. This definition retains the coefficients; it does not take a quotient by source or target automorphisms.

Theorem 3.1 (The marked chain). *Put $K_a = \prod_{j=1}^a (b, c^j)$. Then*

$$T_a = \text{Bl}_{K_a} S_a. \quad (3.2)$$

This is a smooth surface obtained by blowing up the origin and then $a-1$ times the intersection of the newest exceptional curve with the strict transform of $b=0$. Its exceptional divisors E_i have $v_{E_i}(b) = i$, $v_{E_i}(c) = 1$ and form a chain of projective lines. An open cover consists of the first and last charts

$$T^{(0)} = \text{Spec } k[b, k_0], \quad c = bk_0; \quad T^{(a)} = \text{Spec } k[c, w_a], \quad b = c^a w_a,$$

and the intermediate charts

$$T^{(i)} = \text{Spec } k[e_i, h_i], \quad c = e_i h_i, \quad b = e_i^{i+1} h_i^i \quad (1 \leq i < a). \quad (3.3)$$

They are actual charts of the full blow-up, with adjacent transitions

$$\begin{aligned} k_0 &= e_1^{-1}, & b &= e_1^2 h_1, \\ e_{i+1} &= h_i^{-1}, & h_{i+1} &= e_i h_i^2 \quad (1 \leq i < a-1), \\ c &= e_{a-1} h_{a-1}, & w_a &= h_{a-1}^{-1}. \end{aligned} \quad (3.4)$$

All nonadjacent overlaps are obtained by composing these substitutions. The universal embedded curve over T_a is flat and retains the original fixed target X_0 .

3.1. A finite evaluation calculation. Let $d = a + 1$ and $m_a = a(a + 1)/2 + 1$. Gotzmann's regularity theorem gives a closed Grassmannian embedding of the projective Hilbert scheme in degree $m \geq m_a$ [3]. Indeed the Gotzmann representation of $dl + 1$ has d linear binomial terms and $(d - 1)(d - 2)/2$ constant terms, hence m_a terms. We use the ordinary Hilbert scheme after the Segre embedding. The fixed quadratic Segre relations lie in every ideal; the relevant quotient therefore factors through $H^0(X_0, \mathcal{O}(m, m))$, without invoking multigraded regularity.

The restriction matrix M has $q = (a + 1)m + 1$ rows and $N = (m + 1)\binom{m+2}{2}$ columns. The column indexed by $0 \leq l, h, l + h \leq m, 0 \leq t \leq m$ is the monomial $b^h c^l$ in row

$$n = t + a(m - h - l) + (a - 1)l,$$

and is zero in every other row. It is the evaluation of a basis monomial of bidegree (m, m) . Over $b \neq 0$ it is surjective, since restriction to the graph is $H^0(\mathbb{P}^1, \mathcal{O}((a + 1)m))$ in a regularity degree. Its full minors are Plücker coordinates of the Hilbert point.

Proposition 3.2 (Content and row factorization). *For every $m \geq m_a$ put*

$$C = \frac{(a - 1)m(m + 1)}{2}, \quad E_j = m - j + 1 \quad (j < a), \quad E_a = \frac{(m - a + 1)(m - a + 2)}{2}.$$

Then the ideal of maximal minors is exactly

$$I_q(M) = b^C J_{a,m}, \quad J_{a,m} = \prod_{j=1}^a (b, c^j)^{E_j}. \quad (3.5)$$

The divisorial content is b^C , whereas $J_{a,m}$ and all its powers are integrally closed. Changing m changes the weights C, E_j , but not the graph (3.2).

Proof. A nonzero full minor chooses one column from each row; columns from different rows are distinct. Thus $I_q(M)$ is the product of the row ideals Q_n . Put $d_0 = am - n$. Rows with $n \geq am$ have $Q_n = (1)$. For $d_0 \geq 0$ set

$$h_0 = \max\left(0, \left\lceil \frac{d_0 - m}{a - 1} \right\rceil\right), \quad r_0 = d_0 - ah_0.$$

For a fixed h , the least feasible l is $\max(0, d_0 - ah)$, and feasibility is equivalent to $h \geq h_0$ until this minimum is reached. The minimal monomials of Q_n are therefore $b^{h_0+t} c^{\max(0, r_0-at)}$, with $0 \leq t \leq \max(0, \lceil r_0/a \rceil)$. Consequently $Q_n = (b^{h_0})$ if $r_0 \leq 0$, and if $r_0 = av + j > 0$, $0 \leq j < a$, then

$$Q_n = b^{h_0} (b, c^a)^v (b, c^j),$$

where the last factor is omitted when $j = 0$.

The sum of h_0 over the rows is $(a - 1)(1 + \dots + m) = C$. A positive residual value $r_0 \leq m$ occurs $\min(a, m - r_0 + 1)$ times. One occurrence has $h_0 = 0$; the others are specified by $m - a + 2 - h_0 \leq r_0 \leq m - h_0$. The total exponents of the residual factors are

$$\sum_{\substack{1 \leq r_0 \leq m \\ r_0 \equiv j \pmod{a}}} \min(a, m - r_0 + 1) = m - j + 1 \quad (1 \leq j < a),$$

and

$$\sum_{r_0=1}^m \min(a, m - r_0 + 1) \lfloor r_0/a \rfloor = \sum_{n=a}^m (n - a + 1) = E_a.$$

For the first count, write $n = r_0 + t$, $0 \leq t < a$, and count each $n = j, \dots, m$ once. For the second, at fixed n sum $\lfloor r_0/a \rfloor$ for $\max(1, n - a + 1) \leq r_0 \leq n$; the result is $\max(0, n - a + 1)$. This proves (3.5).

For a fixed integer exponent of b , the smallest exponent of c in $J_{a,m}$ is obtained by choosing b first in factors with largest j . Each integer abscissa of its lower Newton polygon is thus attained by an actual generator. The successive integral slopes are $-a, \dots, -1$ with positive horizontal lengths $E_a, \dots, E_1$. All lattice points above the polygon belong to the ideal. Replacing E_j by nE_j proves the same assertion for every power, hence integral closure by the monomial Newton-polyhedron criterion [2, Proposition 1.4.6]. There is no common height-one factor of $J_{a,m}$, since it contains powers of both b and c . Thus b^C is precisely the divisorial content. $\square$

Proof of Theorem 3.1. For an integral affine base, the closure of a rational projective map with coordinates $f_0, \dots, f_N$ is the blow-up of their ideal. On the open where its i th homogeneous coordinate is nonzero the coordinate ring is $A[f_j/f_i]$ inside $\text{Frac}(A)$; these are exactly the affine charts of the Rees Proj. Applying this to the full minors gives the closure of the Hilbert map, since both the Grassmannian and the Hilbert scheme are closed in the chosen projective space and contain its generic image.

Multiplication in degree n by b^{nC} gives a graded isomorphism $\mathcal{R}(J_{a,m}) \simeq \mathcal{R}(b^C J_{a,m})$. Therefore the principal content does not change Proj, though it does change the raw degree- m Fitting ideal. The all-power integral-closure result makes the Rees algebra normal. Its fan has, in clockwise order, rays

$$(0, 1), (1, 1), (2, 1), \dots, (a, 1), (1, 0).$$

The same is true for K_a , by the identical argument with all $E_j = 1$. Each adjacent determinant has absolute value one, so these normal blow-ups are the same smooth surface. Inserting the rays in order gives the stated sequence of point blow-ups. The dual bases on consecutive cones give exactly (3.3) and the end charts; their Laurent identities give (3.4). Thus the cover is exhaustive. Finally pull back the universal Hilbert family along the graph isomorphism. This proves flatness and keeps the embedding in X_0 , not merely the abstract isomorphism class of a curve. $\square$

Proposition 3.3 (The marked principalization category). *Fix the dense coefficient torus $U = \{bc \neq 0\} \subset S_a$. Objects are integral locally Noetherian S_a -schemes V for which U_V is dense and every ideal $(b, c^j)\mathcal{O}_V$ is invertible as a nonzero fractional ideal. Morphisms are arbitrary S_a -morphisms between these objects, with the induced maps on U_V . Then T_a is terminal in this category. For normal objects after any finite flat coefficient cover, the corresponding normalized blow-up is terminal.*

Proof. A product of nonzero fractional ideals is invertible if and only if all its factors are invertible. To check the nontrivial direction locally, if $I_1 \cdots I_a = (t)$, then $I_i(t^{-1} \prod_{j \neq i} I_j) = \mathcal{O}$; this gives a fractional inverse. The blow-up universal property applied to K_a gives a unique map to T_a . After a finite flat cover the same property gives a map to the pulled-back blow-up. On a normal test, pull back its finite normalization and close the generic section over U_V . This is a finite integral birational scheme over the test and hence is the test itself. The lift is unique by density and separatedness. No identification of an arbitrary test with U is required; its specified structural map on U_V is retained. $\square$

4. TANGENT COEFFICIENT DIVISORS

We now apply the conductor identity to a coefficient change which is not diagonal in the given parameters. Throughout this section k is algebraically closed of characteristic zero. Let $a \geq 2$, and let $T_a \rightarrow \text{Spec } k[b, c]$ be the Hilbert graph of

$$[s_0 : t_0] \longmapsto [t_0^a : bs_0^a : cs_0 t_0^{a-1}].$$

The construction and proof of the chain model are given in Section 3; in particular, its exceptional divisors E_i have valuations $v_{E_i}(b) = i$, $v_{E_i}(c) = 1$, for $1 \leq i \leq a$. For integers $r, s \geq 2$ put

$$b = u^r, \quad c = u + v^s. \quad (4.1)$$

Set

$$Y_a = T_a \times_{\text{Spec } k[b,c]} \text{Spec } k[u, v], \quad Z_a = Y_a^\nu.$$

The two reduced divisors $u = 0$ and $u + v^s = 0$ have intersection multiplicity s at the origin. Thus this is not a change by units of two transverse boundary coordinates. We do not replace v by an s th root of $c - u$ on the original coefficient surface.

Theorem 4.1 (The conductor for tangent covers). *The morphism (4.1) is finite flat of degree rs , with Jacobian ideal $(u^{r-1}v^{s-1})$. The graph Y_a is integral and a local complete intersection. For every i there is exactly one prime divisor $D_i \subset Z_a$ over E_i . Its ramification index e_i , residue degree f_i , and valuations are as follows:*

$$i \leq r: \quad g_i = \gcd(rs, i), \quad (v_{D_i}(u), v_{D_i}(v), e_i, f_i) = \left(\frac{si}{g_i}, \frac{i}{g_i}, \frac{rs}{g_i}, g_i \right), \quad (4.2)$$

$$i \geq r: \quad g_i = \gcd(r, is), \quad (v_{D_i}(u), v_{D_i}(v), e_i, f_i) = \left(\frac{is}{g_i}, \frac{r}{g_i}, \frac{rs}{g_i}, g_i \right). \quad (4.3)$$

The two formulas agree at $i = r$. The full conductor on Z_a is

$$c_{Z_a/Y_a} = \mathcal{O}_{Z_a} \left(- \sum_{i=1}^a k_i D_i \right), \quad k_i = \begin{cases} ((rs-1)i - rs) / \gcd(rs, i) + 1, & i \leq r, \\ ((r-1)is - r) / \gcd(r, is) + 1, & i \geq r. \end{cases} \quad (4.4)$$

No other conductor component or additional condition in codimension two occurs. The generic normalization defect over E_i is $f_i k_i / 2$. These assertions hold for all a, r, s in the stated range, not only for monomial arcs in the coefficient surface.

Proof. The intermediate extension defined by $u^r = b$, followed by $v^s = c - u$, is free with basis $u^j v^l$, $0 \leq j < r$, $0 \leq l < s$. The total algebra is $k[u, v]$, and its fraction field has degree rs over $k(b, c)$. Differentiating gives the determinant $rsu^{r-1}v^{s-1}$. The graph and local complete-intersection claims follow from Theorem 2.1 and the smoothness of T_a .

Fix i , write $w = b/c^i$, and localize at the generic point of E_i . The residue field is $k(w)$, with w transcendental. At a prime of the normalization above this DVR let $\alpha = v(u)$, $\beta = v(v)$, and $e = v(c)$. Then

$$r\alpha = ie, \quad v(u + v^s) = e. \quad (4.5)$$

The fundamental equality for the finite normalization of a DVR is $\sum ef = rs$. We shall obtain lower bounds on a single e and f whose product is already rs ; this also proves uniqueness of the prime, rather than assuming it from a Newton polygon.

Suppose first that $i < r$. Then $\alpha < e$, so cancellation in $u + v^s$ forces $s\beta = \alpha$ and $\overline{u/v^s} = -1$. If $g = \gcd(rs, i)$, the valuation equalities force e to be a positive multiple of rs/g . The unit character $v^{rs/g}/c^{i/g}$ has residue η with $\eta^g = (-1)^r w$. The polynomial $T^g - (-1)^r w$ is irreducible over $k(w)$, for example by the w -adic Eisenstein criterion. Hence $f \geq g$. The fundamental equality now forces $e = rs/g$, $f = g$, and a single prime above E_i . This proves (4.2) for $i < r$.

If $i = r$, the residue $\zeta = \overline{u/c}$ satisfies $\zeta^r = w$. It has degree r and is not 1. Since $v^s = c(1 - u/c)$, one has $s\beta = e$. Consequently $e \geq s$ and $f \geq r$, and again equality and uniqueness follow. Both displayed formulas give $(\alpha, \beta, e, f) = (s, 1, s, r)$.

If $i > r$, then $\alpha > e$, so $s\beta = e$ and $\overline{c/v^s} = 1$. Set $g = \gcd(r, is)$. The relations force $e \geq rs/g$, and the residue of $u^{r/g}/v^{is/g}$ has g th power w . The same irreducibility and degree argument gives $e = rs/g$, $f = g$ and (4.3).

The conductor coefficient is now $(r-1)\alpha + (s-1)\beta - e + 1$ by Theorem 2.1. Substitution yields (4.4). The theorem is an equality of ideals, so no extra intersection condition is being inferred merely from these orders. Away from the exceptional locus the pullback is the normal plane $\text{Spec } k[u, v]$. The defect formula follows from Corollary 2.4. $\square$

The residue degrees in this theorem describe extensions of function fields of divisors. They do not count prime divisors. Over a general nonzero closed residue, algebraic closedness and characteristic zero turn the indicated power maps into g_i distinct closed lifts. Special points require the local models, which we compute next in an all-order nonmonomial case.

4.1. The complete square-tangency model. For the rest of this section take $r = s = 2$. We describe the full normal surface and its scheme fibre, not just its divisorial valuations.

Theorem 4.2 (Square tangency in every contact order). *For $b = u^2$, $c = u + v^2$, the surface Z_2 is covered by*

$$B_0 = k[u, v, k_0]/(u + v^2 - u^2 k_0), \quad (4.6)$$

$$B_1 = k[e, z, v]/(v^2 - ez(z-1)), \quad (4.7)$$

$$B_2 = k[c, q, v]/(v^2 - c(1-q)). \quad (4.8)$$

The maps to $\text{Spec } k[u, v]$ are the evident one on B_0 , $u = ez$ on B_1 , and $u = cq$ on B_2 . The overlaps are

$$k_0 = e^{-1}, \quad z = uk_0; \quad c = ez^2, \quad q = z^{-1}.$$

There are precisely two singular points, both of type A_1 ; in the middle chart they are $(e, z, v) = (0, 0, 0)$ and $(0, 1, 0)$. The conductors on these charts are respectively

$$B_0, \quad eB_1, \quad cB_2. \quad (4.9)$$

For every $a > 2$, the normal surface Z_a is obtained from Z_2 by $a-2$ successive ordinary blow-ups at the specified smooth endpoint of the exceptional chain. Its fibre over $(u, v) = (0, 0)$ is a reduced nodal chain of a projective lines, with no embedded points. The two A_1 points persist and no new singularities occur. For every $a \geq 2$,

$$\mathfrak{c}_{Z_a/Y_a} = \mathcal{O}_{Z_a}(-2D_2 - 3D_3 - \cdots - aD_a). \quad (4.10)$$

In particular D_1 and D_2 have the same values $(v(u), v(v)) = (2, 1)$ but different values 4 and 2 on $u + v^2$. A fan in the two original monomial weights cannot distinguish them.

Proof. The three charts of T_2 in Section 3 give raw pullback rings

$$\begin{aligned} A_0 &= k[u, v, k_0]/(u + v^2 - u^2 k_0), \\ A_1 &= k[e, h, u, v]/(u^2 - e^2 h, u + v^2 - eh), \\ A_2 &= k[u, v, w]/(u^2 - (u + v^2)^2 w). \end{aligned}$$

They cover the entire pullback because they are base changes of an open cover of the blow-up. Their integrality has already been proved by finite flatness, so all fractional expressions below lie in their single fraction field.

The ring A_0 is smooth: where $u \neq 0$ the derivative with respect to k_0 is nonzero, and where $u = 0$ the derivative with respect to u is 1. For A_1 adjoin $z = u/e$, integral because $z^2 = h$. Eliminating u, h gives B_1 . For A_2 put $c = u + v^2$ and adjoin $q = u/c$, integral because $q^2 = w$. This gives B_2 . These extensions are finite and birational. The two displayed hypersurfaces are smooth except at the points specified in the theorem; their isolated singularities and the hypersurface S_2 property prove normality. At $z = 0$ and $z = 1$ the factors $z-1$ and z , respectively, are units, so completion and a unit change of e give $v^2 = ez$ or $v^2 = e(z-1)$. These are A_1

singularities. The point $c = v = 0, q = 1$ of B_2 is the same second point. The fractional identities on overlaps prove the stated gluing of the finite normalizations.

We verify the conductors at the intersections directly. Over $D = k[e, h]$, the normal ring B_1 is free with basis $1, z, v, zv$, using $z^2 = h$ and $v^2 = e(h - z)$. The subring A_1 is exactly the free sublattice with basis $1, ez, v, ezv$. Thus an element $A + Bz + Cv + Dzv$ belongs to A_1 if and only if B, D are divisible by e . Requiring its product with z also to belong to A_1 forces A, C to be divisible by e . These necessary conditions say that the element is in eB_1 , and they are sufficient because $eB_1 \subset A_1$ and eB_1 is a B_1 -ideal. This proves the middle equality in (4.9). Over $k[c, w]$, use the basis $1, q, v, qv$ with $q^2 = w$ and $v^2 = c(1 - q)$. The sublattice is $1, cq, v, cqv$; the identical argument gives cB_2 . On the overlap $c/e = z^2$ is a unit, so these are the same ideal. On the first overlap e is a unit, giving B_0 . This is an explicit intersection calculation, independent of any extension of a generic divisor formula.

The complete fibre is equally explicit:

$$B_0/(u, v) = k[k_0], \quad B_1/(u, v) = k[e, z]/(ez), \quad B_2/(u, v) = k[q]. \quad (4.11)$$

For the last equality, $cq = 0$ and $c(1 - q) = 0$ imply $c = 0$. The overlaps compactify the two affine axes of the middle node into two projective lines. These rings are reduced, pure one-dimensional, and have no embedded points. Notice that the second singularity of the ambient surface lies at a smooth point of the fibre.

It remains to prove the assertion for all a , rather than infer it from low-order pictures. The morphism $T_{j+1} \rightarrow T_j$ blows up the last point with coordinates $(c, w_j) = (0, 0)$, where $w_j = b/c^j$. Its inverse image on Z_j is a unique smooth endpoint with local parameters v, q_j . In a neighbourhood of this point put

$$H_j = 1 - v^{j-2}q_j, \quad c = \frac{v^2}{H_j}, \quad u = cv^{j-2}q_j, \quad w_j = q_j^2 H_j^{j-2}. \quad (4.12)$$

For $j = 2$ these are the last chart above, localized at $q_2 = 0$; $H_2 = 1 - q_2$ is a unit. The pullback of (c, w_j) is (v^2, q_j^2) . Its normalized blow-up is the ordinary blow-up of (v, q_j) : the integral closure of $(v^2, q_j^2)^n$ is $(v, q_j)^{2n}$ for every n , by its two-dimensional Newton polygon, and Proj of this Veronese algebra is the ordinary blow-up.

For completeness this computes the required normalized base change, not a substitute birational model. The finite morphism $Z_j \rightarrow T_j$ is flat. Indeed Z_j is a normal surface and hence Cohen–Macaulay, whereas T_j is smooth; a system of parameters of the latter is a regular sequence on the former, giving local freeness over the regular local ring. Flat base change of the blow-up is therefore the blow-up of the pulled-back ideal. The finite birational map from this pullback to Y_{j+1} induces the same function field, so their normalizations agree by transitivity of integral dependence.

On the last chart of the ordinary blow-up set $q_j = vq_{j+1}$. Formula (4.12) becomes the same formula with $j + 1$, proving the induction and uniqueness of the next centre. The old fibre ideal at that centre is $(u, v) = (v)$, so its pullback is the reduced total transform of one smooth component. The blow-up adds one projective line and one node, not an embedded point or a multiple component. The centre is disjoint from the two A_1 points; thus no singularities are added. Finally Theorem 4.1 gives $k_1 = 0$ and $k_i = i$ for $i \geq 2$, as well as the valuation comparison for D_1, D_2 . This proves every assertion. $\square$

Corollary 4.3 (One Hilbert limit, different normalization germs). *For every $a \geq 2$, two coefficient arcs of the square-tangency family can have the same closed coefficients and the same embedded Hilbert limit, while their normalization lifts have nonisomorphic completed local rings: one is regular and the other has an A_1 singularity. Both arcs are generically base-point-free as projective maps.*

Proof. On D_2 in the middle chart the old Hilbert coordinate is $h = z^2$. The points $z = 1$ and $z = -1$ therefore map to the same point $e = 0, h = 1$ of T_2 , and hence give the same embedded curve. The former is an A_1 point of Z_2 , whereas the latter is smooth. They are untouched by the later endpoint blow-ups.

For the point $z = -1$ take $v = \tau, e = \tau^2/2, u = -\tau^2/2$. For the point $z = 1$ take

$$z = 1 + \tau, \quad v = \tau, \quad e = \frac{\tau}{1 + \tau}, \quad u = \tau.$$

Both satisfy $v^2 = ez(z - 1)$ over $k[[\tau]]$ and have $b = u^2, c = u + v^2$. In both cases u is nonzero over $k((\tau))$. If the homogeneous source coordinate s_0 is nonzero, the second coordinate $u^2 s_0^a$ is nonzero; if $s_0 = 0$, the first coordinate t_0^a is nonzero. Thus the entire projective map, including source infinity, is base-point-free on the generic fibre. The regular and A_1 completed local rings have different embedding dimensions, so they cannot be isomorphic. $\square$

5. DIAGONAL COVERS AND THE COMPLETE SCHEME FIBRE

The diagonal calculation provides a second application and a comparison with the tangent case. It also isolates the local algebra needed to pass from generic multiplicities to a complete scheme fibre. Let $b = u^\rho, c = v^\sigma$, where ρ, σ are positive integers, and let $Y = T_a \times_{S_a} \text{Spec } k[u, v]$, $Z = Y^\nu$. Put

$$g_i = \gcd(\rho, i\sigma), \quad n_i = (A_i, B_i) = \left(\frac{i\sigma}{g_i}, \frac{\rho}{g_i} \right), \quad e_i = \frac{\rho\sigma}{g_i}, \quad \ell_i = \min(A_i, B_i). \quad (5.1)$$

Set $n_0 = (0, 1)$ and $n_{a+1} = (1, 0)$, in clockwise order.

Theorem 5.1 (Diagonal parameter boundary). *The surface Y is integral and a local complete intersection. Its normalization is the normal toric surface with consecutive cones $\langle n_i, n_{i+1} \rangle$. The scheme fibre F of $Z \rightarrow \text{Spec } k[u, v]$ over the origin is Cohen–Macaulay of pure dimension one, without embedded points. Its reduction is a nodal chain of a projective lines D_i . The generic length on D_i is ℓ_i , and the nilradical has exact order $\max_i \ell_i$ (order one means it is zero). The conductor is*

$$\mathfrak{c}_{Z/Y} = \mathcal{O}_Z \left(- \sum_{i=1}^a \kappa_i D_i \right), \quad \kappa_i = \frac{\rho i \sigma - \rho - i \sigma}{g_i} + 1. \quad (5.2)$$

There is one divisor over each E_i , with ramification index e_i and residue degree g_i . In particular Y is normal if and only if $\rho = 1$.

5.1. The raw charts and the graded semigroup. Flat base change of the exhaustive cover of T_a gives the rings

$$A_0 = k[u, v, k_0]/(v^\sigma - u^\rho k_0), \quad (5.3)$$

$$A_i = k[u, v, e, h]/(u^\rho - e^{i+1} h^i, v^\sigma - e h) \quad (1 \leq i < a), \quad (5.4)$$

$$A_a = k[u, v, w]/(u^\rho - v^{a\sigma} w). \quad (5.5)$$

The middle ring is finite free over $k[e, h]$, with basis $u^j v^l$, $0 \leq j < \rho, 0 \leq l < \sigma$. The end rings are similarly the pullbacks of the end charts. They inject into the single generic field $k(u, v)$ by finite flatness over the respective coordinate domains of T_a . Thus these equations have neither a hidden vertical component nor an embedded component. They cover Y , not just a surface birational to Y .

For the exponents E_j of Proposition 3.2 put

$$I = \prod_{j=1}^a (u^\rho, v^{j\sigma})^{E_j}, \quad \eta_i = \sum_{j=1}^a E_j \min(i, j).$$

Let $H \subset \mathbb{Z}^3$ be the semigroup generated by $(1, 0, 0)$, $(0, 1, 0)$ and $(\alpha, \beta, 1)$ for the monomial generators $u^\alpha v^\beta$ of I . Its algebra is the ordinary Rees algebra. The group generated by H is all $\mathbb{Z}^3$: subtracting the first two coordinate vectors from any degree-one generator gives $(0, 0, 1)$ in the group. The rational cone of H has the complete half-space description

$$\alpha, \beta, n \geq 0, \quad i\sigma\alpha + \rho\beta \geq n\rho\sigma\eta_i \quad (1 \leq i \leq a). \quad (5.6)$$

Indeed its degree-one section is the Minkowski sum of the Newton segments of the factors plus the nonnegative quadrant; each compact edge has inward normal $(i\sigma, \rho)$ and positive length. The coordinate and degree facets account for its noncompact faces. The normalization is therefore

$$\overline{\mathcal{R}(I)} = k[u^\alpha v^\beta Q^n : (\alpha, \beta, n) \in \mathbb{Z}^3 \text{ satisfying (5.6)}]. \quad (5.7)$$

To justify the grading in this assertion, a lattice point in the cone is a rational nonnegative combination of the listed generators. Multiplying by a common denominator puts a positive multiple in H , so its monomial is integral, satisfying a monic equation $T^N - u^{N\alpha} v^{N\beta} Q^{Nn}$. Conversely an integral monomial lies in the cone by every facet valuation. Thus the saturation is $\text{Cone}(H) \cap \mathbb{Z}^3$ with its unchanged third-coordinate grading, not a claim that the original semigroup was saturated in a fixed degree. This is the standard affine semigroup-normalization theorem; see [2, Chapters 1 and 5].

The raw evaluation ideal is $u^{\rho C} I$. The degree-preserving isomorphism $fQ^n \mapsto u^{n\rho C} fQ^n$ identifies its Rees algebra with $\mathcal{R}(I)$ and, on their fraction fields, identifies the normalizations by $Q \mapsto u^{\rho C} Q$. This distinguishes a graded presentation from its Proj. Formula (5.7) gives the fan in Theorem 5.1.

5.2. A coordinate ideal on an arbitrary proper cone.

Lemma 5.2 (The full coordinate ideal). *Let l, h be clockwise primitive rays spanning a two-dimensional cone strictly smaller than the nonnegative quadrant, and lying in that quadrant. In the character lattice $M = \mathbb{Z}^2$ put*

$$A = k[\{m \in M : \langle l, m \rangle, \langle h, m \rangle \geq 0\}].$$

The characters $u = \chi^{(1,0)}$, $v = \chi^{(0,1)}$ belong to A , and

$$(u, v)A = \bigoplus_{\substack{m \in M : \langle l, m \rangle \geq \min(l_1, l_2) \\ \langle h, m \rangle \geq \min(h_1, h_2)}} k\chi^m. \quad (5.8)$$

It is the intersection of the two height-one symbolic valuation ideals with these orders. In particular it is reflexive, and $A/(u, v)A$ has no zero-dimensional associated prime whenever it is nonzero.

Proof. Negative exponents are allowed exactly when both dual-cone inequalities hold. They are characters regular on this chart, although they need not be polynomials in the original plane. Divisibility by u means $m - (1, 0)$ satisfies both inequalities; divisibility by v means $m - (0, 1)$ does. The forward inclusion in (5.8) follows immediately.

If $l_1 \leq l_2$ and $h_1 \leq h_2$, the character v/u is regular, so $(u, v) = (u)$ and the reverse inclusion follows. If both inequalities are reversed, use $(u, v) = (v)$. Equality of one pair of coordinates causes no difficulty in these cases.

In the remaining case $l_1 < l_2$ and $h_1 > h_2$. Write $m = (x, y)$, $L = x + y$. We first prove $L \geq 1$ for an integral m on the right of (5.8). If $L \leq 0$, the low-ray inequality $l_1 L + (l_2 - l_1)y \geq l_1$ forces $y \geq 0$, with $y > 0$ if $l_1 > 0$; the high-ray inequality $h_1 L - (h_1 - h_2)y \geq h_2$ forces $y \leq 0$, with $y < 0$ if $h_2 > 0$. At least one of l_1, h_2 is positive, because the full quadrant is excluded. This is a contradiction.

Subtraction of $(1, 0)$ can fail only at the high ray: the low threshold is already l_1 . Subtraction of $(0, 1)$ can fail only at the low ray: the high threshold is already h_2 . If both fail, then

$$l_1 L + (l_2 - l_1)y < l_2, \quad h_1 L - (h_1 - h_2)y < h_1.$$

Since $L \geq 1$ and y is integral, the first inequality gives $y \leq 0$, while the second gives $y > 0$. This contradiction proves that one of u, v divides χ^m in A .

In a normal affine toric ring the two ray valuations are the height-one valuations of its invariant prime divisors. The monomial ideal on the right is their symbolic-ideal intersection; orders at all other height-one primes are nonnegative automatically, since each character is a unit on the coefficient torus. Consequently it is the divisorial ideal specified by these orders and is reflexive. The normal surface ring is S_2 , and a reflexive ideal has depth two at each closed point. In $0 \rightarrow (u, v)A \rightarrow A \rightarrow A/(u, v)A \rightarrow 0$, the depth lemma therefore gives depth at least one for the quotient at every closed point in its support. All nonempty support has dimension one, so there is no zero-dimensional associated prime. $\square$

Proof of the fibre assertions in Theorem 5.1. Every maximal cone in the fan is proper because $a \geq 2$. Apply Lemma 5.2 to obtain, on the whole surface,

$$(u, v)\mathcal{O}_Z = \mathcal{O}_Z \left(- \sum_i \ell_i D_i \right).$$

This is an equality of actual ideals. It proves purity, Cohen–Macaulayness and the generic lengths by localization at the DVR of D_i . The radical is the divisorial ideal with order one on each D_i . If $L = \max \ell_i$, every ordinary product of L sections of the radical has order at least L on all D_i and hence belongs to $(u, v)\mathcal{O}_Z$. This proves the upper bound on the ordinary power of the nilradical, without asserting that an ordinary product of reflexive ideals is reflexive. At a component with $\ell_i = L > 1$, the quotient of a DVR by its L th uniformizer power has nonzero $(L - 1)$ st nilradical power, proving exactness.

At a crossing choose the standard finite cyclic cover of the toric chart. If its adjacent determinant is Δ , its ring is $k[x, y]^{\mu_\Delta}$, with weights coprime to Δ in each coordinate. The reduced boundary union is the invariant ring of $k[x, y]/(xy)$, since invariants are exact in characteristic zero. The latter invariant ring is

$$k[x^\Delta, y^\Delta]/(x^\Delta y^\Delta).$$

Equivalently it is $k[x^\Delta] \times_k k[y^\Delta]$. Thus the completed node ring is $k[[X, Y]]/(XY)$ even if the ambient toric surface is singular. At end charts only one boundary axis is in the fibre. The exceptional orbit closures are projective lines by the complete one-dimensional star fans, so they glue to the claimed chain. $\square$

5.3. The conductor and its extension across intersections. The diagonal map has Jacobian ideal $(u^{\rho-1}v^{\sigma-1})$. Over E_i , the primitive normalized valuation is n_i ; its ramification index is e_i in (5.1). The residue character $\zeta = u^{\rho/g_i}/v^{i\sigma/g_i}$ satisfies $\zeta^{g_i} = b/c^i$. This polynomial is irreducible over $k(b/c^i)$, so there is one divisor with residue degree g_i ; $e_i g_i = \rho\sigma$. Theorem 2.1 now gives

$$(\rho - 1)A_i + (\sigma - 1)B_i - e_i + 1 = \frac{\rho i \sigma - \rho - i \sigma}{g_i} + 1,$$

including every intersection as an equality of conductor ideals. For reference, the following general extension argument also explains why a separate hidden punctual condition is impossible here.

Lemma 5.3 (*S_2 extension of conductor conditions*). *Let A be an S_2 Noetherian domain with finite normalization B . Then $A = \bigcap_{\text{ht } P=1} A_P$ in its fraction field, and*

$$(A : B) = B \cap \bigcap_{\text{ht } P=1} (A_P : B \otimes_A A_P). \quad (5.9)$$

When the local conductor is written by valuations in the semilocal normalization of A_P , the intersection includes each of its maximal ideals, namely the primes of B above P .

Proof. For the first assertion take a fraction z in all A_P and consider the finite module $M = (A + Az)/A$. If it is nonzero, a minimal prime of its support has height at least two. Localize there; M then has finite length, A has depth at least two, and the torsion-free module $A + Az$ has depth at least one. The exact sequence

$$0 \longrightarrow A \longrightarrow A + Az \longrightarrow M \longrightarrow 0$$

and the depth inequality $\text{depth } M \geq \min(\text{depth } A - 1, \text{depth}(A + Az))$ give depth at least one, contrary to nonzero finite length. Thus $M = 0$. Testing $zb \in A$ for each $b \in B$ gives (5.9).

Localization of finite normalization gives a semilocal normal one-dimensional ring over A_P . Its maximal ideals are not to be identified with P itself; all primes above P must be used. Every nonzero conductor ideal of this semilocal Dedekind ring is specified by its orders in these DVRs. Intersecting their conditions inside B proves the assertion, with no additional condition in codimension two. In the geometric applications all rings are excellent, so computing these orders after faithfully flat completion is legitimate; in the present paper they were computed directly by trace and valuation. $\square$

The coefficients in (5.2) vanish when $\rho = 1$. If $\rho > 1$, the last is positive, since $a\sigma \geq 2$: for $R, Q \geq 2$, $RQ - R - Q + \gcd(R, Q) > 0$. This proves the normality criterion. Its asymmetry records the orientation of the chain $\prod_j (b, c^j)$, not a symmetry-breaking assertion about arbitrary covers of the plane.

If $\rho = \sigma = d$, the conductor generators on the first and every intermediate normalized chart are u^{d-1} , and on the last chart $v^{a(d-1)}$. Indeed their valuations are the conductor coefficients on all rays of that chart, including order zero on its nonexceptional axis; the ideal identity follows from the theorem. At an intermediate intersection the same character is used on both sides. At the last intersection $u = v^a \zeta$, where ζ is a unit, so the generators are related by the unit ζ^{d-1} . This verifies their gluing, not just equality of divisorial orders on one chart.

5.4. Quotient conventions and an unequal-cover chart. To specify the quotient type without an orientation ambiguity, order the two primitive rays r_1, r_2 counterclockwise and put $\Delta = \det(r_1, r_2) > 0$. Extend r_1 to an oriented lattice basis (r_1, t) and write $r_2 = pr_1 + \Delta t$. Then the chart has cyclic type $\frac{1}{\Delta}(-p, 1)$, where p is read modulo Δ and $\gcd(p, \Delta) = 1$. This follows because $t = (-pr_1 + r_2)/\Delta$ acts on the smooth covering characters with these weights. Interchanging the two rays interchanges the weights; changing t changes p by a multiple of Δ .

Example 5.4 (An unequal cover at a quotient crossing). Take $(a, \rho, \sigma) = (3, 2, 3)$. On the first raw chart

$$A = k[u, v, \theta]/(v^3 - u^2\theta)$$

the normalization is the cubic Veronese ring

$$B = k[s^3, s^2t, st^2, t^3] = k[u, v, w, \theta]/(v^2 - uw, vw - u\theta, w^2 - v\theta).$$

The map is $u = s^3$, $v = s^2t$, $w = st^2$, $\theta = t^3$. The extra generator $w = v^2/u$ is integral, and the Veronese ring is normal (it is the invariant ring for the scalar μ_3 action). It has the same fraction field and is finite, so is the normalization. The cone rays are $(0, 1)$, $(3, 2)$ and the quotient type is $\frac{1}{3}(1, 1)$.

Here $B = A + Aw$, $w^2 = v\theta$, and $uw = v^2$, $vw = u\theta$. Thus $(u, v)B$ is contained in the conductor. Conversely the valuation of w on the exceptional ray is one, while u, v, θ have values 3, 2, 0. The

conductor coefficient is two by (5.2); every invariant monomial of order at least two is divisible by u or v , since $w^2 = v\theta$. Consequently

$$\mathfrak{c}_{B/A} = (u, v)B, \quad B/(u, v) = k[w, \theta]/(w^2).$$

This displays both the conductor and the nonreduced full fibre at the quotient point. The fibre has generic length two and no embedded associated prime. On the complete surface the three conductor orders are 2, 3, 8, while the three generic fibre lengths are 2, 1, 2; these are different invariants.

5.5. Marked logarithmic compatibility. For the diagonal family the normal fan is the coarsest subdivision on which each $\min(\rho p, j\sigma q)$ is linear. Equivalently it is terminal among normal integral tests with dense coefficient torus which make every $(u^\rho, v^{j\sigma})$ invertible, with the category specified as in Proposition 3.3. For ordered transverse Cartier divisors U, V on a smooth surface, these ideals are intrinsically $\mathcal{O}(-\rho U) + \mathcal{O}(-j\sigma V)$. Unit changes of local equations preserve them; normalized blow-ups consequently glue. Subsequent root covers compose by integral closure in the same function field. The conductor-divisor transitivity is the special case of Theorem 2.5, which also applies when the coefficient change is not diagonal.

If ∂Z is the full reduced toroidal boundary and $\partial S'$ is the union of the coordinate axes, then

$$K_Z + \partial Z = \pi^*(K_{S'} + \partial S').$$

Both sides are Cartier (indeed represented by the divisor of the logarithmic torus form); Z itself is simplicial, so its invariant Weil divisors are $\mathbb{Q}$ -Cartier. A toric resolution preserves this equality and has a simple normal crossing reduced boundary, proving log canonicity of the pair. This is a statement about the parameter space, not stable or semistable reduction of its universal embedded curves.

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